

AI Application Lecture 6

Probabilistic Language Models and Sampling

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^aI thank my collaborator, Jing WANG, for cooperating in creating the teaching materials.

Outline

1. Introduction
2. Preliminaries: Mathematical Notations
3. How do we use a NN in discrete token generations? - Probabilistic language model
4. How do we determine the output token from a distribution? - Sampling and beam search
5. Can we adapt the generation to real application preference? - Hyperparameters
6. Summary
7. Next Time

Review of the previous lecture

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- We learned that **tokenization** is the crucial entry and exit point for natural language processing pipelines, converting between raw text strings and sequences of tokens.
- We examined Byte Pair Encoding (BPE), understanding its motivation: balancing sequence length reduction against the need to handle unknown words.

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- **Premise:** We now know how to convert text to and from token sequences.
- **Question:** How do we make a neural network generate a probable sequence of tokens?

A neural network is a continuous function, but a token sequence is a discrete object.

We will have the neural network represent a **probabilistic language model** and use that model along with a pseudo-random number generator to build a **token generator**.

Learning outcomes for this lecture

By the end of this lecture, you should be able to:

- Understand the framework of viewing a continuous neural network as a **probabilistic language model** by using softmax normalization to obtain a probability mass function.

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By the end of this lecture, you should be able to:

- Understand the framework of viewing a continuous neural network as a **probabilistic language model** by using softmax normalization to obtain a probability mass function.
- Recursively construct a **token generator** from a probabilistic language model and a pseudo-random number sequence.
- Rigorously define and generate token sequences using various samplers, including greedy search, beam search, and methods involving temperature, top- k , top- p , and repetition penalty.

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Preliminaries: mathematical notations

Set:

- Sets: $\mathcal{A}$
- Membership: $x \in \mathcal{A}$
- Empty set: $\{\}$
- Roster notation: $\{a, b, c\}$
- Set-builder: $\{x \in \mathcal{A} | P(x)\}$
- Cardinality: $|\mathcal{A}|$
- Real numbers: $\mathbb{R}, \mathbb{R}_{>0}, \mathbb{R}_{\geq 0}$
- Integers: $\mathbb{Z}, \mathbb{Z}_{>0}, \mathbb{Z}_{\geq 0}$
- Integer range: $[1, k]_{\mathbb{Z}} := \{1, \dots, k\}$

Function:

- $f : \mathcal{X} \rightarrow \mathcal{Y}$: f maps from $\mathcal{X}$ to $\mathcal{Y}$.
- $y = f(x)$: The output of f for input x .

Definition:

- (LHS) $:=$ (RHS): Left side is defined by the right side.

Preliminaries: mathematical notations

Sequence: Denoted by $\mathbf{a} = (a_1, a_2, \dots)$.

- A function $\mathbf{a} : [1, n]_{\mathbb{Z}} \rightarrow \mathcal{A}$.
- Length is denoted by $|\mathbf{a}|$.

Vector: Denoted by $\mathbf{v}$.

- A column of numbers, $\mathbf{v} \in \mathbb{R}^n$.
- i -th element is v_i .

Matrix: Denoted by $\mathbf{A}$.

- $m \times n$ matrix: $\mathbf{A} \in \mathbb{R}^{m,n}$.
- (i, j) -th element is $a_{i,j}$.
- Transpose: $\mathbf{A}^{\top}$.

Tensor: Denoted by $\underline{\mathbf{A}}$.

- Simply a multi-dimensional array.
- Vector $\rightarrow$ 1st-order, Matrix $\rightarrow$ 2nd-order.

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Question: A neural network is a continuous function. In natural language generation, we want to output a one-hot vector sequence.

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How do we use a neural network to output one-hot vectors?

Motivation

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However, the desired output in natural language processing (a token sequence) is a discrete object.

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Solution: Instead of outputting the token sequence itself, the neural network outputs **scores for tokens**. These scores are then converted into probabilities via the softmax function, defining a **probabilistic language model**.

Terminologies about tokens

Let's formalize our terms.

Definition (Vocabulary and Token Sequences)

The set of all possible values a token can take is called the **vocabulary**, denoted by $\mathcal{V}$.

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- The set of token sequences of length n is $\mathcal{V}^n$.
- The set of all token sequences of any finite length is $\mathcal{V}^* = \mathcal{V}^0 \cup \mathcal{V}^1 \cup \mathcal{V}^2 \cup \dots$.

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Definition (Probabilistic Language Model (Most General Form))

Let a finite vocabulary $\mathcal{V} = \{1, 2, \dots, D\}$ be fixed. A **probabilistic language model** is a function $P(\cdot \mid \cdot)$ such that, given any finite-length token sequence $t = (t_1, \dots, t_n) \in \mathcal{V}^n$, $P(\cdot \mid t)$ is a probability mass function on $\mathcal{V}$.

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More formally, for any $\mathbf{t} \in \mathcal{V}^*$, the following must hold:

$$\sum_{v \in \mathcal{V}} P(v \mid \mathbf{t}) = 1, \quad P(v \mid \mathbf{t}) \geq 0 \quad (\text{for all } v \in \mathcal{V}) \quad (1)$$

Remark on the probabilistic language model definition

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It also specifies the joint distribution of an entire sequence via the chain rule of probability:

$$P((t_1, \dots, t_n)) = \prod_{i=1}^n P(t_i | (t_1, \dots, t_{i-1}))$$

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Constructing the joint distribution directly is possible but harder to adapt for arbitrary-length outputs compared to modeling conditional probabilities.

Construction from a neural network

To construct a probabilistic language model from a neural network, we must solve two issues:

- **Handling Variable-Length Inputs:** How can a fixed neural network architecture process token sequences of different lengths?

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- **Handling Variable-Length Inputs:** How can a fixed neural network architecture process token sequences of different lengths?
- **Handling Probability Constraints:** How do we ensure the network's output is a valid probability distribution (non-negative and sums to 1)?

Handling variable-length inputs

To handle variable-length inputs, two approaches exist:

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- **Giant Neural Network Approach:** Use a huge network that accepts a very long input of fixed length L . Pad shorter inputs to match this length.

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- **Giant Neural Network Approach:** Use a huge network that accepts a very long input of fixed length L . Pad shorter inputs to match this length.
- **Variable-Length Input Neural Network Model Approach:** Use an architecture with parameter sharing and recursive structures (like RNNs or Transformers). This allows¹ constructing a network $f_{\theta}^{(n)}$ tailored to the input length n from a single set of parameters θ .

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Most successful approaches (CNNs, RNNs, Transformers) take the second direction.

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A crucial point is that we don't need to store an infinite number of networks $f_{\theta}^{(n)}$ for all possible lengths n .

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Instead, we only need an **algorithm** that can mechanically construct the specific network $f_{\theta}^{(n)}$, once the input length n is known.

This algorithm takes the input length, n , as an argument and dynamically constructs the appropriate network architecture for that specific length. All we need to store permanently is the fixed, finite-dimensional parameter vector θ .

Handling probability constraints

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The standard method is **softmax normalization**.

Definition (Construction via Softmax Normalization)

Given a variable-length input function $f_{\theta}^{(\cdot)}$ that maps a sequence of length n to a vector of scores in $\mathbb{R}^{|\mathcal{V}|}$, $f_{\theta}^{(n)} : \mathcal{V}^n \rightarrow \mathbb{R}^{|\mathcal{V}|}$.

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The probability of token v given history t is defined as:

$$P_{f_{\theta}^{(\cdot)}}(v \mid t) := \frac{\exp\left(f_{\theta,v}^{(|t|)}(t)\right)}{\sum_{u \in \mathcal{V}} \exp\left(f_{\theta,u}^{(|t|)}(t)\right)} \quad (2)$$

The resulting function $P_{f_{\theta}^{(\cdot)}}(\cdot \mid \cdot)$ is a probabilistic language model.

Handling probability constraints

In practice, we often introduce a hyperparameter called **temperature** ($T > 0$) to control the shape of the probability distribution.

Definition (Construction with Temperature)

The probability is defined by dividing the scores by T before applying softmax:

$$P_{f_{\theta}^{(\cdot)}, T}(v \mid \mathbf{t}) := \frac{\exp\left(\frac{1}{T} \cdot f_{\theta, v}^{(|\mathbf{t}|)}(\mathbf{t})\right)}{\sum_{u \in \mathcal{V}} \exp\left(\frac{1}{T} \cdot f_{\theta, u}^{(|\mathbf{t}|)}(\mathbf{t})\right)} \quad (3)$$

Handling probability constraints

Remark

Changing the temperature T ² controls the "peakedness" of the distribution.

- As $T \downarrow 0$ (low temperature): The distribution becomes sharper. Differences in scores are exaggerated, making the model more confident and deterministic. The probability mass concentrates on the highest-scoring token.

²The name "temperature" originates from statistical mechanics (the Boltzmann distribution, a probability distribution obtained by applying softmax to the product of the inverse of temperature and the negative energy corresponding to $f_{\theta,u}^{(|t|)}(t)$ in the previous definition, is fundamental in statistical mechanics).

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- When $T = 1$, we recover the original softmax definition.

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Neural network as a probabilistic language model - answer

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Answer: We do not use the neural network's output directly. Instead, we regard the output as a probability distribution of the output.

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How can we generate a **sequence** of tokens?

Two basic ideas

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- **Sampling (Stochastic):** At each step, randomly draw the next token according to the probability distribution $P(\cdot | t)$. This introduces diversity.
- **Selecting the Maximum (Deterministic):** At each step, simply pick the token with the highest probability. This leads to more consistent but potentially repetitive outputs.

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- A PRNG uses a **deterministic algorithm** and an initial **seed** to produce a sequence of numbers that appears random (long period, high uniformity).
- A famous example is the **Mersenne Twister**³ [3].

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Definition: pseudo-random sequences and token generators

First, let's define a pseudo-random sequence.

Definition (Pseudo-random sequence)

Given a fixed initial seed $s \in \mathbb{Z}$, a generator PRNG deterministically returns an infinite sequence of numbers in $[0, 1)$, $\mathbf{u}(s) = (u_1, u_2, \dots)$.

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Now, we can define a token generator.

Definition (Token Generator (General Form))

A **token generator** is a deterministic function $\text{Gen}_{(P, \text{PRNG})}$ that takes an input sequence t_{in} and a seed s , and returns an output sequence t_{out} .

$$\text{Gen}_{(P, \text{PRNG})} : \mathcal{V}^* \times \mathbb{Z} \rightarrow \mathcal{V}^* \quad (4)$$

Internally, it sequentially appends tokens based on the model P and the numbers from $\mathbf{u}(s)$, stopping when a **termination condition** is met.

Definition: pseudo-random sequences and token generators

Remark

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Remark

If the initial seed s is fixed, the output is **completely deterministic**.

The apparent "randomness" comes entirely from the PRNG. The generation rule itself is a deterministic mathematical function.

Sampling

This is the most basic sampling method, also known as inverse transform sampling.

Definition (Sampling using a Probabilistic Language Model)

Given an initial sequence $t^{(0)}$, a probabilistic model P , a PRNG, and a stopping condition Stop . For $i = 0, 1, 2, \dots$:

1. If $\text{Stop}(t^{(i)})$ is true, stop and return $t^{(i)}$.

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$$F^{(i)}(v) := \sum_{u \leq v} P(u | t^{(i)}).$$

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4. Find the next token: $t_{i+1} := \min\{v \in \mathcal{V} \mid F^{(i)}(v) > u_{i+1}\}.$

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5. Append the token: $\mathbf{t}^{(i+1)} := (\mathbf{t}^{(i)}, t_{i+1})$ and repeat.

Sampling

Example (Two Steps of Naive Sampling)

Let $\mathcal{V} = \{1, 2, 3\}$, input $t_{\text{in}} = ()$, and stop at length 2. The probability distributions are:

$$P(\cdot \mid ()) = (0.50, 0.30, 0.20)$$

$$P(\cdot \mid (1)) = (0.30, 0.40, 0.30)$$

$$P(\cdot \mid (2)) = (0.10, 0.20, 0.70)$$

$$P(\cdot \mid (3)) = (0.60, 0.10, 0.30)$$

The PRNG gives us $u_1 = 0.68$, $u_2 = 0.72$. Let's find the output.

Sampling

Step 1:

- Current sequence $t^{(0)} = ()$.
- The probability distribution is $P(\cdot|()) = (0.50, 0.30, 0.20)$.
- The cumulative distribution (CDF) is $F^{(0)} = (0.50, 0.80, 1.00)$.
- Our first random number is $u_1 = 0.68$.
- We look for the smallest token v where $F^{(0)}(v) > 0.68$.
 - $F^{(0)}(1) = 0.50 \not> 0.68$
 - $F^{(0)}(2) = 0.80 > 0.68$
- So, the next token is $t_1 = 2$. New sequence $t^{(1)} = (2)$.

Sampling

Step 2:

- Current sequence $t^{(1)} = (2)$.
- The probability distribution is $P(\cdot|(2)) = (0.10, 0.20, 0.70)$.
- The CDF is $F^{(1)} = (0.10, 0.30, 1.00)$.
- Our second random number is $u_2 = 0.72$.
- We look for the smallest token v where $F^{(1)}(v) > 0.72$.
 - $F^{(1)}(1) = 0.10 \not> 0.72$
 - $F^{(1)}(2) = 0.30 \not> 0.72$
 - $F^{(1)}(3) = 1.00 > 0.72$
- So, the next token is $t_2 = 3$. New sequence $t^{(2)} = (2, 3)$.

The length is now 2, so we stop. The final output is $(2, 3)$.

Sampling

Exercise (Manual Calculation Exercise (Naive Sampling))

Let $\mathcal{V} = \{1, 2, 3\}$, start with $()$, and stop at length 2.

$$P(\cdot \mid ()) = (0.40, 0.40, 0.20)$$

$$P(\cdot \mid (1)) = (0.25, 0.25, 0.50)$$

$$P(\cdot \mid (2)) = (0.10, 0.70, 0.20)$$

$$P(\cdot \mid (3)) = (0.50, 0.30, 0.20)$$

Given $u_1 = 0.41$, $u_2 = 0.20$, find the output sequence.

Sampling

Answer

Step 1:

- $t^{(0)} = ()$.
- $P(\cdot | ()) = (0.40, 0.40, 0.20)$.
- CDF: $(0.40, 0.80, 1.00)$.
- $u_1 = 0.41$. Since $0.40 < 0.41 \leq 0.80$, we select $t_1 = 2$.
- New sequence: $t^{(1)} = (2)$.

Sampling

Answer

Step 2:

- $t^{(1)} = (2)$.
- $P(\cdot|(2)) = (0.10, 0.70, 0.20)$.
- CDF: $(0.10, 0.80, 1.00)$.
- $u_2 = 0.20$. Since $0.10 < 0.20 \leq 0.80$, we select $t_2 = 2$.
- New sequence: $t^{(2)} = (2, 2)$.

Stop. The output sequence is $(2, 2)$.

Let's now turn to the deterministic approach.

Selecting the maximum probability element

The simplest deterministic method is greedy search.

Definition (Greedy decoding)

Given an initial sequence $\mathbf{t}^{(0)}$ and a stopping condition Stop. At each step i :

1. Select the token with the highest probability:

$$t_{i+1} := \arg \max_{v \in \mathcal{V}} P \left(v \mid \mathbf{t}^{(i)} \right)$$

(Ties are broken by a fixed rule, e.g., lowest token ID).

2. Append the token: $\mathbf{t}^{(i+1)} := (\mathbf{t}^{(i)}, t_{i+1})$.
3. Repeat until Stop is met.

Selecting the maximum probability element

Example (Greedy Search (Two Steps, Numerical))

Let $\mathcal{V} = \{1, 2, 3\}$, stop at length 2. Probabilities:

$$P(\cdot \mid ()) = (0.55, 0.40, 0.05)$$

$$P(\cdot \mid (1)) = (0.20, 0.70, 0.10)$$

- **Step 1:** Given $()$, the probabilities are $(0.55, 0.40, 0.05)$. The max is at index 1. So, $t_1 = 1$.
- **Step 2:** Given (1) , the probabilities are $(0.20, 0.70, 0.10)$. The max is at index 2. So, $t_2 = 2$.

The output is $(1, 2)$.

Limitations of greedy search and beam search

Greedy search does not guarantee finding the sequence with the highest overall joint probability.

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Greedy search does not guarantee finding the sequence with the highest overall joint probability.

The joint probability of a sequence is $P(t_1, t_2) = P(t_1 | ()) \cdot P(t_2 | (t_1))$. A locally optimal choice at step 1 might lead to a globally suboptimal sequence.

Example (Greedy Fails)

Let $\mathcal{V} = \{A, B\}$, stop at length 2.

$$P(\cdot | ()) = (0.60, 0.40)$$

$$P(\cdot | (A)) = (0.50, 0.50)$$

$$P(\cdot | (B)) = (0.95, 0.05)$$

Limitations of greedy search and beam search

Greedy Search Path:

- Step 1: $P(A|()) = 0.6 > P(B|()) = 0.4$. Choose A.
- Step 2: $P(A|(A)) = 0.5$. Choose A (tie-break).
- Result: (A, A) . Joint probability: $P(A, A) = 0.6 \times 0.5 = 0.30$.

Limitations of greedy search and beam search

Greedy Search Path:

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- Result: (A, A) . Joint probability: $P(A, A) = 0.6 \times 0.5 = 0.30$.

True Best Sequence: Let's check all possibilities:

- $P(A, A) = 0.6 \times 0.5 = 0.30$
- $P(A, B) = 0.6 \times 0.5 = 0.30$
- $P(B, A) = 0.4 \times 0.95 = \mathbf{0.38}$
- $P(B, B) = 0.4 \times 0.05 = 0.02$

The sequence (B, A) has the highest probability, but greedy search missed it.

Beam search

Beam search partially addresses the issue of greedy search.

Beam search

Beam search partially addresses the issue of greedy search.

Idea: Instead of keeping only the single best choice at each step, keep track of the B most probable partial sequences (the "beam").

Definition (Beam Search with beam width B)

1. Start with an empty sequence.
2. At each step i :
 - For each of the B sequences currently in the beam, generate all $|\mathcal{V}|$ possible next sequences and calculate their probabilities.
 - From this set of $B \cdot |\mathcal{V}|$ candidates, select the top B sequences with the highest overall probabilities. These form the new beam.
3. Repeat until a stop condition is met, then return the highest-scoring sequence from the final beam.

Beam search

Let's re-run the previous example with beam search, using a beam width $B = 2$.

Step 1:

- Initial candidates: (A) with probability 0.6 and (B) with probability 0.4.
- Since $B = 2$, we keep both.
- Beam $\mathcal{B}_1 = \{(A, 0.6), (B, 0.4)\}$.

Beam search

Let's re-run the previous example with beam search, using a beam width $B = 2$.

Step 1:

- Initial candidates: (A) with probability 0.6 and (B) with probability 0.4.
- Since $B = 2$, we keep both.
- Beam $\mathcal{B}_1 = \{(A, 0.6), (B, 0.4)\}$.

Step 2:

- Expand from A : (A, A) with probability $0.6 \cdot 0.5 = 0.3$. And (A, B) also with probability 0.3.
- Expand from B : (B, A) with probability $0.4 \cdot 0.95 = 0.38$. And (B, B) with probability 0.02.
- Candidates' probabilities: $\{0.30, 0.30, 0.38, 0.02\}$.
- The top 2 are (B, A) and (A, A) (or (A, B)). The best is (B, A) .

Generating a sequence - answer

Question: How can we generate a **sequence** of tokens?

Answer:

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Answer: We iteratively select the maximum probability token or stochastically sample a next token according to the probability distribution determined by the neural network.

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Question: Since we do not directly use the neural network to determine the output, we have room for adapting the token generation rule so that it is preferable in applications.

What

Question: Since we do not directly use the neural network to determine the output, we have room for adapting the token generation rule so that it is preferable in applications.

What modifications we can apply to the token generation algorithm, without changing the neural network itself?

Modifying probabilistic language models

In practice, the raw probability distribution from the model is often modified before sampling.

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Motivations:

- **Preventing Repetition:** Avoid generating monotonous, repetitive text.

Modifying probabilistic language models

In practice, the raw probability distribution from the model is often modified before sampling.

Motivations:

- **Preventing Repetition:** Avoid generating monotonous, repetitive text.
- **Excluding Low-Score Tokens:** Avoid generating nonsensical text by completely ruling out tokens that the model considers very unlikely.

Modifying probabilistic language models

Here are some popular techniques:

- **Repetition Penalty [2]:** Decrease the scores of tokens that have already appeared in the generated sequence.

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- **Repetition Penalty [2]:** Decrease the scores of tokens that have already appeared in the generated sequence.
- **Temperature:** As we've seen, lowering the temperature T makes the distribution sharper and less random.
- **Top-k Sampling:** Consider only the top k most probable tokens and set the probability of all others to zero.

Modifying probabilistic language models

Here are some popular techniques:

- **Repetition Penalty [2]:** Decrease the scores of tokens that have already appeared in the generated sequence.
- **Temperature:** As we've seen, lowering the temperature T makes the distribution sharper and less random.
- **Top-k Sampling:** Consider only the top k most probable tokens and set the probability of all others to zero.
- **Top-p (Nucleus) Sampling [1]:** Consider the smallest set of most probable tokens whose cumulative probability exceeds a threshold p . This set is called the "nucleus."

Modifying probabilistic language models

These modifications are often applied in a specific order. Let's look at the common pipeline used by libraries like Hugging Face.

Definition (Construction of Conditional PMF in Hugging Face)

Given scores $s_{|t}$ for history t :

1. Apply **repetition penalty**.
2. Apply **temperature** scaling.
3. Apply **top- k** filtering.
4. Apply **softmax** to get probabilities $\pi(v)$.
5. Apply **top- p** (nucleus) filtering and renormalize.

Modifying probabilistic language models

Let's look closer at the repetition penalty step.

repetition penalty: Let $H(t)$ be the set of previously seen tokens and $\lambda > 0$ be the penalty factor.

$$s_{v|t}^{\text{rep}} \leftarrow \begin{cases} s_{v|t} & \text{if } v \notin H(t), \\ \frac{1}{\lambda} s_{v|t} & \text{if } v \in H(t) \text{ and } s_{v|t} \geq 0, \\ \lambda s_{v|t} & \text{if } v \in H(t) \text{ and } s_{v|t} < 0. \end{cases}$$

4

⁴This operation is mathematically unnaturally complex. Simply subtracting a constant from the scores would be simpler and would provide the same operation for the same probability mass function, regardless of any constant bias in the score function. However, in practice, the operation defined here is used and is said to be useful. The definition of this operation (multiplicative rather than additive modification) seems to originate from [2].

Modifying probabilistic language models

Example (Hugging Face Style Sampling)

Let $\mathcal{V} = \{1, 2, 3, 4\}$, start with $()$, and stop at length 2. Scores:

$$\mathbf{s}_{|()} = (2.0, 1.0, 0.0, -1.0)$$

$$\mathbf{s}_{|(1)} = (1.5, 0.0, 0.5, -0.2)$$

$$\mathbf{s}_{|(2)} = (2.5, 0.5, 0.5, -0.5)$$

$$\mathbf{s}_{|(3)} = (0.5, 0.0, 1.0, -1.6)$$

$$\mathbf{s}_{|(4)} = (0.3, 0.0, 0.0, -2.2)$$

Hyperparameters: $T = 0.5$, $\lambda = 1.2$, $k = 3$, $p = 0.9$. PRNG gives $u_1 = 0.50$, $u_2 = 0.82$. Find the output.

Modifying probabilistic language models

Step 1 (History $t = ()$, Seen tokens $H(()) = \{\}$)

1. **(rep)** No tokens seen, scores unchanged: $(2.0, 1.0, 0.0, -1.0)$.
2. **(temp)** Divide by $T = 0.5$ (i.e., multiply by 2): $(4.0, 2.0, 0.0, -2.0)$.
3. **(top- k)** Keep top $k = 3$ scores. 4th score becomes $-\infty$: $(4.0, 2.0, 0.0, -\infty)$.
4. **(softmax)** Probabilities $\pi \approx (0.8668, 0.1173, 0.0159, 0)$.
5. **(top- p)** Cumulative sum is $(0.8668, 0.9841, \dots)$. For $p = 0.9$, nucleus is $\{1, 2\}$.
Renormalize: $\tilde{\pi} \approx (0.8808, 0.1192, 0, 0)$.
6. **(sample)** CDF is $(0.8808, 1.0)$. $u_1 = 0.50 < 0.8808 \implies t_1 = 1$.

Modifying probabilistic language models

Step 2 (History $t = (1)$, Seen tokens $H((1)) = \{1\}$)

1. **(rep)** Score for token 1: $1.5/\lambda = 1.5/1.2 = 1.25$. Scores: $(1.25, 0.0, 0.5, -0.2)$.
2. **(temp)** Divide by $T = 0.5$: $(2.5, 0.0, 1.0, -0.4)$.
3. **(top- k)** Keep top $k = 3$ scores: $(2.5, 0.0, 1.0, -\infty)$.
4. **(softmax)** Probabilities $\pi \approx (0.7662, 0.0629, 0.1710, 0)$.
5. **(top- p)** Sorted probs: $(0.7662_1, 0.1710_3, 0.0629_2)$. Cum sum: $(0.7662, 0.9372, \dots)$. Nucleus is $\{1, 3\}$. Renormalize: $\tilde{\pi} \approx (0.8176, 0, 0.1824, 0)$.
6. **(sample)** CDF is $(0.8176, 0.8176, 1.0, 1.0)$. $u_2 = 0.90$. Since $0.8176 < 0.82 \leq 1.0$, we select $t_2 = 3$.

Final output: $(1, 3)$.

Exercise

Example (Hugging Face Style Sampling)

Use the same setting. Let $\mathcal{V} = \{1, 2, 3, 4\}$, start with $()$, and stop at length 2.

Scores:

$$\mathbf{s}_{|()} = (2.0, 1.0, 0.0, -1.0)$$

$$\mathbf{s}_{|(1)} = (1.5, 0.0, 0.5, -0.2)$$

$$\mathbf{s}_{|(2)} = (2.5, 0.5, 0.5, -0.5)$$

$$\mathbf{s}_{|(3)} = (0.5, 0.0, 1.0, -1.6)$$

$$\mathbf{s}_{|(4)} = (0.3, 0.0, 0.0, -2.2)$$

Hyperparameters: $T = 0.5$, $\lambda = 1.2$, $k = 3$, but no top- p (or, $p = 1$). PRNG gives $u_1 = 0.50$, $u_2 = 0.82$. Find the output.

Exercise - answer

Step 1 (History $t = ()$, Seen tokens $H(()) = \{\}$)

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4. **(softmax)** Probabilities $\pi \approx (0.8668, 0.1173, 0.0159, 0)$.
5. **(top- p)** The top- p is not applied. $\tilde{\pi} \approx (0.8668, 0.1173, 0.0159, 0)$.
6. **(sample)** CDF is $(0.8668, 0.9841, 1.0)$. $u_1 = 0.50 < 0.8668 \implies t_1 = 1$.

Exercise - answer

Step 2 (History $t = (1)$, Seen tokens $H((1)) = \{1\}$)

1. **(rep)** Score for token 1: $1.5/\lambda = 1.5/1.2 = 1.25$. **Scores:** $(1.25, 0.0, 0.5, -0.2)$.
2. **(temp)** Divide by $T = 0.5$: $(2.5, 0.0, 1.0, -0.4)$.
3. **(top- k)** Keep top $k = 3$ scores: $(2.5, 0.0, 1.0, -\infty)$.
4. **(softmax)** Probabilities $\pi \approx (0.7662, 0.0629, 0.1710, 0)$.
5. **(top- p)** The top- p is not applied. $\tilde{\pi} \approx (0.7662, 0.0629, 0.1710, 0)$.
6. **(sample)** CDF is $(0.7662, 0.8291, 1.0, 1.0)$. $u_2 = 0.82$. Since $0.7662 < 0.82 \leq 0.8291$, we select $t_2 = 2$.

Final output: $(1, 2)$.

Sampling hyperparameters - answer

Question: What

Sampling hyperparameters - answer

Question: What modifications we can apply to the token generation algorithm, without changing the neural network itself?

Answer:

Sampling hyperparameters - answer

Question: What modifications we can apply to the token generation algorithm, without changing the neural network itself?

Answer: To increase the diversity in generation, we can modify the generation algorithm by changing, e.g., the repetition penalty, temperature, Top-k, Top-p, etc.

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Summary

Let's summarize the key takeaways from today's lecture.

- Neural networks, being continuous functions, are naturally used as **probabilistic language models**. They output scores, which are converted to a probability mass function (PMF) via softmax. The joint probability of a sequence is then defined by the chain rule.

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- We can rigorously define various text generation methods within a unified framework of a **token generator**, which is a deterministic function of a model, a stopping condition, and a seed for a pseudo-random number generator.

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- We can rigorously define various text generation methods within a unified framework of a **token generator**, which is a deterministic function of a model, a stopping condition, and a seed for a pseudo-random number generator.
- This framework includes methods like greedy search, beam search, and various sampling techniques that modify the probability distribution using temperature, top- k , top- p , and repetition penalty.

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- The learning rules for this mapping.
- Its statistical interpretation.

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